

NICHOLAS TIETJE

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EDUCATION

Northeastern University

Master of Science in Computer Science, GPA 3.9/4.0

Bachelor of Science in Chemical Engineering and Computer Science, GPA 3.7/4.0

Boston, MA

Sept 2025 - May 2026

Sept 2021 - May 2025

- **Relevant Coursework:** Distributed Systems, Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Software Engineering, Mobile Application Development, Object Oriented Design, Algorithms & Data Structures, Database Design

EXPERIENCE

Software Engineer Intern

Northeastern University (NExT Consulting)

Boston, MA

Sept 2025 - Dec 2025

- Delivered technical leadership to 6 client-facing software teams, improving React/TypeScript, Python, and testing practices
- Mentored 30+ developers on production engineering workflows including code review, CI/CD, debugging, and system design
- Evaluated code quality, system design, and client communications to maintain high delivery standards for external partners

Software Engineer Intern

Fidelity Investments

Boston, MA

May 2025 - Aug 2025

- Developed request-money & spending-insight features for Fidelity's Kotlin Android app, improving parent visibility into youth account activity
- Streamlined teen user experience by implementing role-based feature gating to dynamically suppress or customize features
- Prototyped Kotlin Multiplatform migration for internal CMS service, evaluating cross-platform business logic & UI feasibility

Software Engineer Intern

Northeastern University (NExT Consulting)

Boston, MA

Sept 2024 - Dec 2024

- Built GitMarks, a full-stack TypeScript/React & Go platform that integrates GitHub workflows into coding assignments
- Gathered requirements from professors & deployed a pilot program across several Northeastern University CS/DS courses
- Integrated GitHub APIs for repository creation, assignment distribution, deadline tracking, submissions, and grade-enabled pull requests with TA feedback

Software Engineer Intern

State Street

Boston, MA

Jul 2024 - Sept 2024

- Designed a Python-based internal notification management system for 3000+ users, supporting subscription management, targeted delivery, and administrative workflows
- Gathered requirements from stakeholders to define API contracts, data models, and compliance-aware access patterns

Undergraduate Researcher

Computational Modeling in Chemical Engineering at Northeastern University

Boston, MA

Feb 2024 - Jul 2024

- Contributed to RMG-Py, a computational modeling software, by developing features & code optimizations in Python
- Migrated RMG-Py output generation to YAML, improving interoperability & reducing output execution time by up to 50%

Teaching Assistant

Northeastern University

Boston, MA

Jan 2024 - May 2024

- Mentored 250+ students in CS 3200 Database Design, delivering support through Piazza discussions & 1-on-1 office hours

PROJECTS

The Wizard's Archive: Collaborative TTRPG Management Platform



TypeScript, React, Convex, Cloudflare Workers, Yjs, BlockNote, Tiptap, TanStack Router, Tanstack Query, Tailwind CSS

- Built a real-time collaborative document platform spanning rich-text notes, canvases, maps, and files with live multi-user conflict-free updates, cross-content wiki-links, and embedded documents
- Designed granular access-control, block-level sharing, view-as-user previews, document snapshots, rollback flows, and collaboration logic backed by 2000+ automated tests

LifeLinked: A MTG Life Tracking App



Kotlin, Compose/Kotlin Multiplatform, Gradle, Ktor, Koin, Android Studio, S3, Lambda

- Launched a cross-platform mobile app, attracting 10,000+ installs across Google Play & App Store
- Achieved 95%+ code sharing between iOS & Android by leveraging Compose Multiplatform for unified UI & business logic

TECHNICAL SKILLS

Languages:

Python, TypeScript/JavaScript, Kotlin, Go, Java, MySQL, PostgreSQL, SQLite, HTML/CSS

Tools/Frameworks:

React, Node.js, Git, Tanstack, Vite, Docker, FastAPI, Android, REST APIs, WebSockets

Cloud/Testing:

AWS, Cloudflare Workers, Terraform, GitHub Actions, Vitest, Playwright, CI/CD, JWT